

German University in Cairo (GUC)
Media Engineering and Technology (MET)

Automated Hallucination Detection Report

Neuro-Symbolic Verification via Knowledge Graphs

Bachelor Thesis Project Output

Mohammed Mahmoud Elnaggar

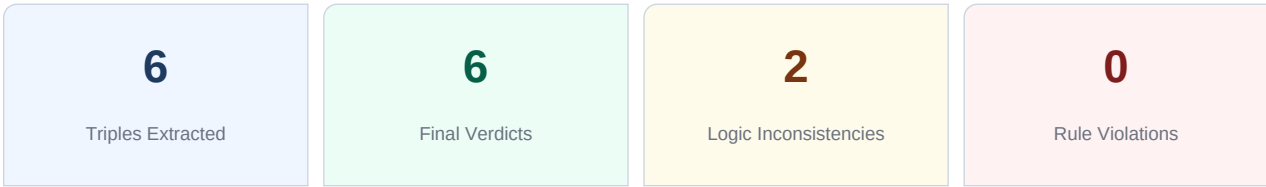
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REPORT OVERVIEW & METHODOLOGY

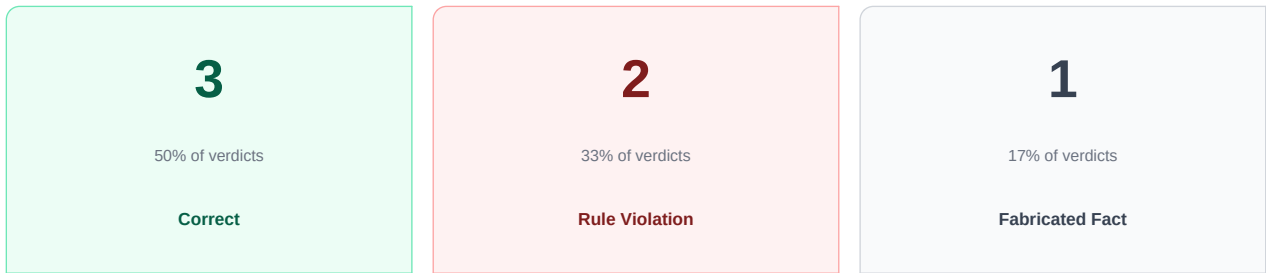
This document outlines the results of the KGVS verification pipeline, a neuro-symbolic approach designed to enforce domain rules and verify factual consistency in Black-Box Large Language Models (LLMs).

Pipeline Stages:

1. **Triple Extraction & Canonicalization:** Unstructured LLM outputs are decomposed into relational triples (Subject, Predicate, Object) and mapped to the formal schema of the underlying database.
2. **Symbolic Constraint Verification:** Triples are subjected to rigorous logical constraint checks to identify domain-rule violations prior to graph traversal.
3. **Graph-based Factual Verification:** Canonicalized triples are queried against the Neo4j Knowledge Graph to categorize them as strictly correct, unverifiable, or explicitly fabricated.



VERDICT DISTRIBUTION



EXTRACTED TRIPLES

All verifiable triples extracted from the LLM response prior to canonicalization or knowledge-graph checking.

#	Subject	Predicate	Object	Type	Predicate Definition
1	300	HAS_APPROVAL_ODDS	8		The approval odds percentage or category associated with the specific credit score.
2	840	HAS_APPROVAL_ODDS	8		The approval odds percentage or category associated with the specific credit score.
3	600	HAS_APPROVAL_ODDS	8		The approval odds percentage or category associated with the specific credit score.
4	555	HAS_APPROVAL_ODDS	36		The approval odds percentage or category associated with the specific credit score.
5	560	HAS_APPROVAL_ODDS	36		The approval odds percentage or category associated with the specific credit score.
6	557	HAS_APPROVAL_ODDS	36		The approval odds percentage or category associated with the specific credit score.

CANONICALIZED TRIPLES

Predicates normalized against the database schema. The Confidence column reflects alignment quality.

Subject	Original Predicate	Object	Canonical Predicate	Conf.
300	HAS_APPROVAL_ODDS	8	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	MEDIUM
840	HAS_APPROVAL_ODDS	8	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	MEDIUM
600	HAS_APPROVAL_ODDS	8	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	MEDIUM
555	HAS_APPROVAL_ODDS	36	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	MEDIUM
560	HAS_APPROVAL_ODDS	36	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	MEDIUM
557	HAS_APPROVAL_ODDS	36	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	MEDIUM

LOGICAL INCONSISTENCIES

Contradictions detected between triples within the LLM response, independent of any external knowledge source.

Label	Type	Severity	Explanation	Triples Involved
logically_inconsistent	MONOTONICITY_VIOLATION	MEDIUM	A higher credit score (600) has a lower approval odds (8) than a lower credit score (555, 557, 560) with approval odds (36), violating the INCREASING direction requirement.	600 -> APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD -> 8 555 -> APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD -> 36
logically_inconsistent	MONOTONICITY_VIOLATION	MEDIUM	A higher credit score (840) has a lower approval odds (8) than a lower credit score (555, 557, 560) with approval odds (36), violating the INCREASING direction requirement.	840 -> APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD -> 8 555 -> APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD -> 36

FACTUALLY INCONSISTENT TRIPLES

All triples labelled factually_inconsistent from the combined verification agent. Part A shows Pass-1 symbolic constraint violations. Part B shows Pass-2 hallucination verdicts with full evidence.

PART A — Symbolic Constraint Violations (Pass 1)

No symbolic constraint violations detected.

PART B — Hallucination Detection Verdicts (Pass 2)

Subject	Predicate	Object	Verdict	Label	Inconsistency Type	Match	Violated	Conf.	Explanation
557	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	36	Fabricated Fact	factually_inconsistent	FABRICATED_FACT	NONE	SUBJECT	0.90	Object+predicate exist in DB but no matching subject found; subject is fabricated.
300	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	8	Correct	factually_consistent	—	STRICT	—	1.00	The object matches exactly the DB candidate value for the subject and predicate. Ev: subject: 300, predicate: APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD, object: 8
840	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	8	Rule Violation	factually_inconsistent	RULE_VIOLATION	NONE	OBJECT	1.00	The object '8' contradicts the stored DB value '97' for the same subject and predicate. Ev: subject: 840, predicate: APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD, object: 97

Subject	Predicate	Object	Verdict	Label	Inconsistency Type	Match	Violated	Conf.	Explanation
600	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	8	Rule Violation	factually_inconsistent	RULE_VIOLATION	NONE	OBJECT	1.00	The object '8' contradicts the stored DB value '41' for the same subject and predicate. Ev: subject: 600, predicate: APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD, object: 41
555	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	36	Correct	factually_consistent	—	STRICT	—	1.00	The object matches exactly the DB candidate value for the subject and predicate. Ev: subject: 555, predicate: APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD, object: 36
560	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	36	Correct	factually_consistent	—	STRICT	—	1.00	The object matches exactly the DB candidate value for the subject and predicate. Ev: subject: 560, predicate: APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD, object: 36

VERDICT LEGEND

Verdict	Description
Correct	All three triple components verified against the knowledge graph.
Rule Violation	One or more components contradict database evidence.
Unverifiable	No usable DB evidence found for any component of this triple.
Predicate Mismatch	Subject entity exists, but the stated predicate is not a valid relation.
Fabricated Fact	Triple asserts a relationship that is in db but the subject does not supported by db.